Fiscal Year:	FY 2016	**Task Last Updated:**	FY 09/28/2015
PI Name:	Govind, Shubha Ph.D.		
Project Title:	Does Spaceflight Alter the Virulence of a Natural Parasite of Drosophila?		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	10031-9101	**Congressional District:**	13
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	10/31/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Taylor, Elizabeth	**Contact Phone:**	650.604.1783
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 10/31/2017 per NSSC information (Ed., 8/3/16)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Bhattacharya, Sharmila Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	NNX15AB42G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The seminal discovery of Toll and Toll-NF-KB signaling as a central mediator of innate immune responses in the model insect Drosophila catalyzed similar discoveries in other organisms including innate immunity in humans. These findings led to an appreciation of the fundamental roles of innate immunity in human immune deficiency on the one hand, and immune health of insects on the other. In Drosophila larvae, primed, but quiescent hematopoietic progenitors are called into action from an organ with functional similarities to mammalian bone marrow. Blood cells signal the fat body (equivalent to the mammalian liver) to activate the expression of antimicrobial peptide genes (humoral immunity). Antimicrobial peptides efficiently combat deadly bacteria and fungi. The Leptopilina wasps are natural metazoan enemies of Drosophila and their attack kills fruit flies. Wasp attack (e.g., L. boulardi) activates both the humoral and cellular immune arms of larval hosts, each being dependent on Toll-NFKB signaling. Previous NASA-funded studies of Drosophila have shown that stresses in the space environment compromise hematopoietic development and modulate larval and adult innate immunity. We hypothesize that stimuli from the space environment will also alter the virulence of the parasitic wasps, and therefore their success on the fly hosts cultured in space. In Drosophila-wasp co-culture experiments using the NASA Fruit Fly Lab hardware over 30-days, we propose to (1) examine whether the fly or the wasp partner wins the arms race in space; (2) examine the effects of spaceflight on the anti-parasite response in fruit fly hosts; and (3) examine if virulence of "space-cultured" wasps is different from counterparts grown on Earth. While both L. boulardi and L. heterotoma parasites always succeed on the wild type, only the latter wasp species produces unique cell-killing activities residing in its virus-like particles. It also suppresses humoral immunity completely. We will examine if changes in the expression of VLP genes underlies changes in parasite virulence affected by spaceflight. Host defense and virulence will be inferred from a suite of other well-established assays of fly/host survival and measures of immune response.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have performed preliminary experiments to compare wasp development at low and high temperatures to optimize the time of development during the mission. The physical container in which the organisms will be cultured is being optimized to suit the needs of the experiments without added burden on the crew. This container will be tested in laboratory experiments. Fruit fly parasites are being cultured in the co-investigator's lab so that the entire team is more knowledgeable about the system.
Task Progress:	During the last reporting period, protocols of rearing fruit fly parasites were modified to optimize the time needed for optimal development of the parasite. Furthermore, significant modifications in the hardware were considered by the team and are in the process of being implemented. The resulting apparatus will be tested in experiments planned for this year. We have performed preliminary experiments to compare wasp development at low and high temperatures to optimize the time of development during the mission. The physical container in which the organisms will be cultured is being optimized to suit the needs of the experiments without added burden on the crew. This container will be tested in laboratory experiments. Fruit fly parasites are being cultured in the co-investigator's lab so that the entire team is more knowledgeable about the system. Three presentations were made in conferences. One publication was submitted and accepted during the last year. Ching-Jung, L., Smibert, P., Zhao, X., Hu, J.F., Ramroop, J., Kellner, S.M., Benton, M., Saint, R., Govind, S., Dedon, P.C., Sternglanz, R. Eric C. Lai, E.C. "An extensive allelic series of Drosophila kae1 mutants reveals diverse and tissue-specific requirements for t6A biogenesis." RNA/2015/053934.
Bibliography Type:	Description: (Last Updated: 01/09/2024)
Abstracts for Journals and Proceedings	Panettieri JG, Govind S. "Discovery of N-Acyl aspartic acid and glutamic acid derivatives in Drosophila melanogaster and first insights into their biological role" Bioactive Lipids in Cancer, Inflammation and Related Diseases Conference 2015, Budapest, Hungary, July 12-15, 2015. Bioactive Lipids in Cancer, Inflammation and Related Diseases Conference 2015, Budapest, Hungary, July 12-15, 2015. , Jul-2015
Abstracts for Journals and Proceedings	Heavner ME, Govind S, Singh S. "Predictive venomics: Using hybrid modeling to gain functional insights into a novel, predicted GTPase from an immune suppressive wasp." Biophysical Society 58th Annual Meeting, San Francisco, CA, November 2014. Biophysical Society 58th Annual Meeting, San Francisco, CA, November 2014. Abstract #L4156. , Nov-2014
Abstracts for Journals and Proceedings	Heavner ME, Singh SM, Govind S. "Venomics: Wasp-induced immune modulation in D. melanogaster." NIH Minority Health and Health Disparities Grantees' Conference, National Harbor, MD, December 2014. NIH Minority Health and Health Disparities Grantees' Conference, National Harbor, MD, December 2014. , Dec-2014